



# CLIMATE–AQI CORRELATION MODULE — TECHNICAL DOCUMENTATION

## 1. MODULE OVERVIEW

### 1.1 Section Context

This module establishes the **analytical relationship between climate variables (temperature, wind) and air quality (AQI)**.

It acts as a **linking analytical layer** between:

Temperature & Wind Module ↔ AQI Module



### Purpose

To quantify how **meteorological conditions influence air pollution levels**, enabling:

- Identification of environmental drivers of AQI fluctuations
- Understanding seasonal and regional effects
- Supporting predictive and policy-relevant insights

## 2. DATA SOURCES & INPUT TABLES



### 2.1 Temperature Data (From Climate Module)

| Source Table                       | Key Fields                  |
|------------------------------------|-----------------------------|
| fact_climate / monthly_temperature | Date, Station_ID, Mean Temp |
| Dim_Stations_All                   | Location metadata           |



### 2.2 AQI Data (From AQI Module)

| Source Table | Key Fields                    |
|--------------|-------------------------------|
| Fact_AQI     | NAPS_ID, Date, Pollutant, AQI |

| Source Table        | Key Fields |
|---------------------|------------|
| Station AQI derived | Daily AQI  |

## ✓ 2.3 Station Mapping

Since datasets come from:

- Climate stations
- NAPS AQI stations

Mapping is done via:

City / Province / Region alignment

## ✓ 2.4 Time Dimension

Shared:

Model Calendar

Used for:

- Month-level aggregation
- Seasonal grouping

# 3. DATA PREPARATION FOR CORRELATION

## 3.1 Aggregation Strategy

Correlation requires **compatible granularity**.

### ✓ AQI Aggregation

From your pipeline:

Daily AQI → Monthly AQI  
`mean_AQI = AVG(AQI per month per city/station)`

## ✓ Temperature Aggregation

Daily Mean Temp → Monthly Mean Temp

## 3.2 Alignment Keys

Final dataset constructed on:

City + Month + Year

## ✓ Resulting Dataset Structure

| Column    | Description      |
|-----------|------------------|
| City      | Location         |
| Province  | Region           |
| Year      | Year             |
| Month     | Month            |
| Mean Temp | Climate variable |
| Mean AQI  | Pollution metric |

## 3.3 Feature Engineering

### ✓ Seasonal Mapping

From your pipeline:

| Month   | Season |
|---------|--------|
| Dec–Feb | Winter |
| Mar–May | Spring |
| Jun–Aug | Summer |
| Sep–Nov | Fall   |



# 4. CORRELATION MODEL SPECIFICATION

## 4.1 Objective

To compute the statistical relationship between:

Climate variable (X) and AQI (Y)

## 4.2 Mathematical Definition

### ✓ Pearson Correlation Coefficient

$$r(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \times \sqrt{\sum (Y_i - \bar{Y})^2}}$$

Where:

- $X$  = temperature (or climate variable)
- $Y$  = AQI
- $r \in [-1, +1]$

## 4.3 Interpretation

| Correlation (r) | Meaning                      |
|-----------------|------------------------------|
| +1              | Strong positive relationship |
| 0               | No relationship              |
| -1              | Strong negative relationship |

## 4.4 Model Implementation (From your notebook)

Grouped computation:

GROUP BY:

City, Month, Pollutant

COMPUTE:

correlation(Mean Temp, AQI)

## 5. OUTPUT TABLE — ClimateAQI\_correlation

### 5.1 Table Structure

From your Power BI screenshot:

| Column           | Description                      |
|------------------|----------------------------------|
| City             | Location                         |
| Province         | Province                         |
| Month            | Month                            |
| Weather_Variable | Climate input (Temp, Snow, etc.) |
| Pollutant        | AQI pollutant                    |
| Correlation      | Computed value                   |
| Season           | Derived                          |

### 5.2 Grain

City × Month × Pollutant

## 6. DATA VALIDATION

### ✓ 6.1 Data Completeness

- Ensure both:
  - temperature values
  - AQI values

exist for each grouping

### ✓ 6.2 Null Handling

Drop rows where:

AQI = NULL OR Temp = NULL

## ✓ 6.3 Consistency Check

- No duplicate keys:

City + Month + Pollutant

# 7. BI MODEL INTEGRATION

## 7.1 Relationship Structure

### ✓ Link with AQI

ClimateAQI\_correlation  
↔ Fact\_AQI

#### Mapping:

City / Province + Month

### ✓ Link with Temperature

ClimateAQI\_correlation  
↔ Climate Tables

## 7.2 Relationship Type

Derived Analytical Relationship (not transactional)

### ✓ Why?

- Data is aggregated
- Used for analysis, not filtering base facts

# 8. ANALYTICAL INTERPRETATION

### ✓ Example Insights

## ✓ Positive Correlation

Higher temperature → Higher AQI

Possible causes:

- Ozone formation increases with heat
- Stagnant air conditions

## ✓ Negative Correlation

Higher temperature → Lower AQI

Possible causes:

- Increased air dispersion
- Reduced particulate accumulation

## ✓ Seasonal Patterns

- Winter:
  - higher particulate pollution
- Summer:
  - ozone-driven AQI spikes

# 9. MODEL LIMITATIONS

## ✓ Temporal Aggregation Bias

- Monthly averages smooth extreme events

## ✓ Station Mapping Approximation

- Climate station ≠ AQI station (not exact location match)

## ✓ Causality Limitation

Correlation ≠ causation

## **Missing Variables**

- Wind direction / dispersion not explicitly included
- Industrial / traffic factors not modeled